# Self Selection

Raffaele Saggio

Labor Economics-EIEF, Summer 2026

June 10, 2026

# Self-selection

- One of (if not THE) most powerful ideas in economics: *Self-selection*
- Individuals choose between opportunities based on heterogeneous unobserved returns
- Applications: occupational choice, immigration, labor force participation, educational choice, treatment choice, any economics subfield...
- We'll start with general discussion of self-selection models and some related econometrics, then look at some applications

# Roy Model

- Roy (1951) sought to understand the influence of occupational choice on the observed distribution of earnings
- Consider individuals indexed by $i$ choosing a binary variable $D_i \in \{0, 1\}$, e.g. hunting vs. fishing
- $Y_{i1}$ and $Y_{i0}$ are $i$'s potential outcomes (e.g. earnings) with and without treatment
- Realized outcome is $Y_i = Y_{i0} + (Y_{i1} - Y_{i0})D_i$
- Pure Roy (1951) model: Individuals want to maximize $Y_i$, so choose the alternative with the best potential outcome:

$$D_i = 1\{Y_{i1} > Y_{i0}\}$$

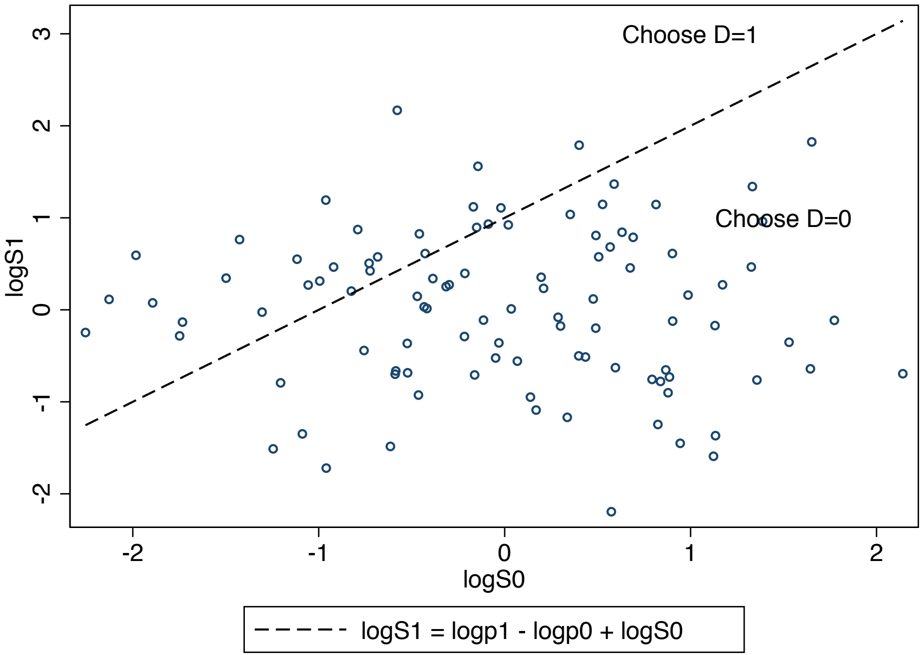
# Roy Model

- Questions of interest:
  - Will the best hunters hunt?
  - Will the best fishermen/women fish?
- Suppose potential outcomes are given by

$$Y_{id} = p_d S_{id}, \;\; d \in \{0, 1\}$$

- $S_{id}$ is skill in occupation $d$, and $p_d$ is price of output
- A worker who is indifferent between the two occupations satisfies

$$\log S_{i1} = \log p_0 - \log p_1 + \log S_{i0}$$

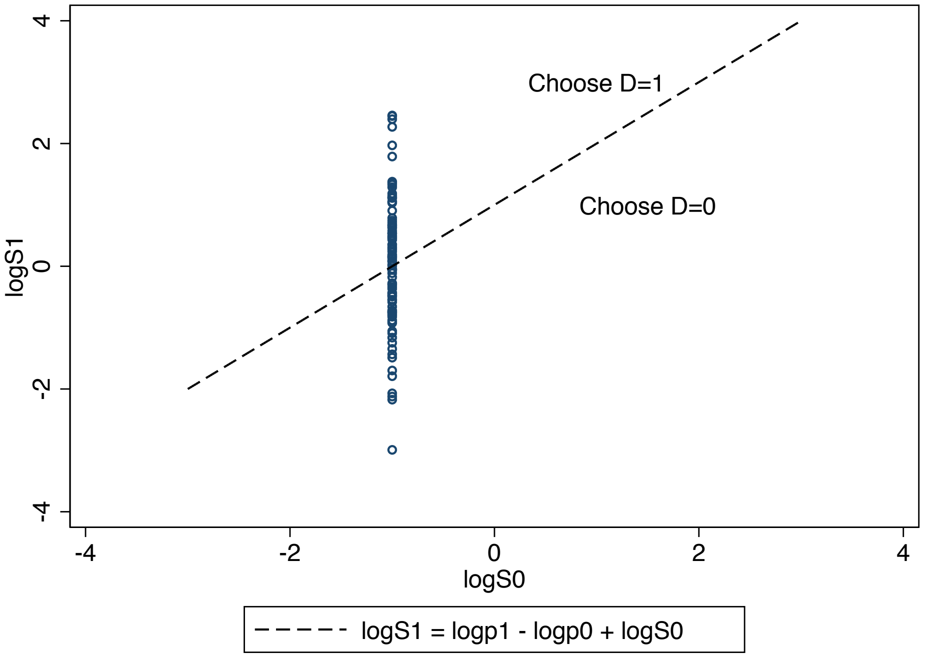

Choose D=1
Choose D=0
logS1
logS0
-2
-1
0
1
2
3
logS1 = logp1 - logp0 + logS0

# Roy Model: Special Cases

- Suppose there is no variation in the non-treated outcome, so $S_{i0} = \bar{S}_0 \ \forall i$
- In this case the decision rule is

$$D_i = 1\left\{S_{i1} \geq \left(\frac{p_0}{p_1}\right)\bar{S}_0\right\}$$

- Those with the most skill in sector 1 choose $D_i = 1$
- Everyone with $D_i = 1$ earns more than anyone with $D_i = 0$



Choose D=1
Choose D=0
logS1
logS0
4
2
0
-2
-4
-4
-2
0
2
4
logS1 = logp1 - logp0 + logS0

# Roy Model: Special Cases

- Suppose we have perfect correlation between $\log S_{i0}$ and $\log S_{i1}$:

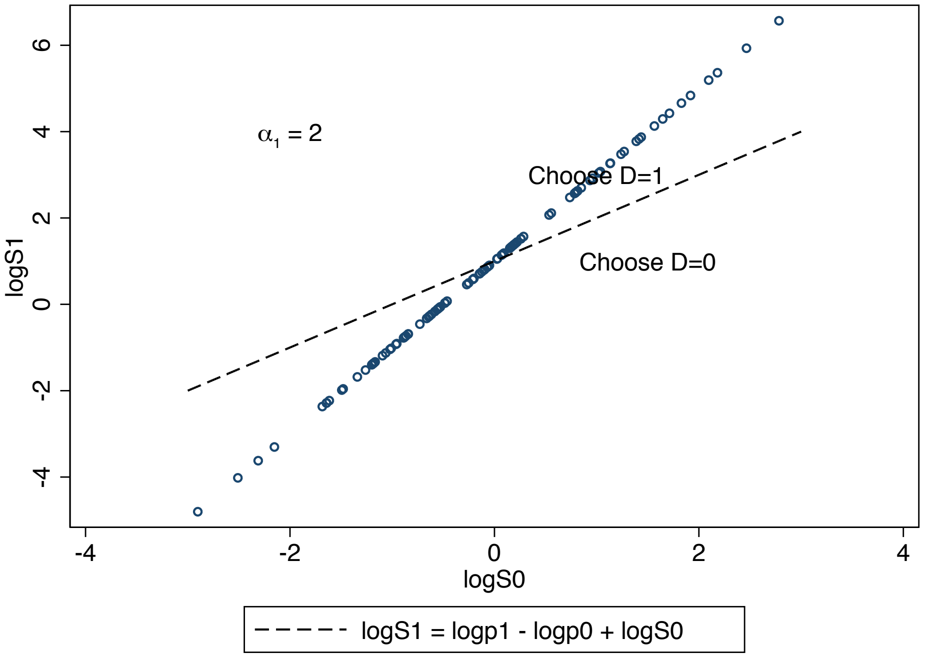
$$\log S_{i1} = \alpha_0 + \alpha_1 \log S_{i0},\ \ \alpha_1 > 0$$

- This is a one-factor model
- Decision rule:

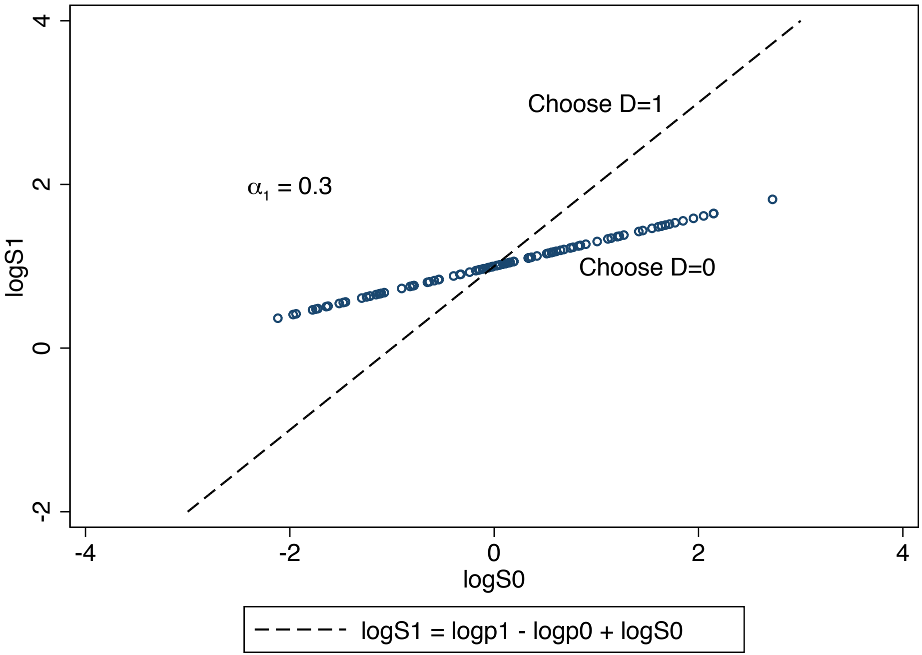
$$D_i = 1\left\{\alpha_0 + \alpha_1 \log S_{i0} \geq \log p_0 - \log p_1 + \log S_{i0}\right\}$$

$$= 1\left\{(\alpha_1 - 1)\log S_{i0} \geq \log p_0 - \log p_1 - \alpha_0\right\}$$

- Higher skilled choose $D_i = 1$ iff $\alpha_1 \geq 1$
- Note that $Var(\log S_{i1}) = \alpha_1^2 Var(\log S_{i0})$ . Higher skilled choose the sector with higher variance of earnings



$\alpha_1 = 2$
Choose D=1
Choose D=0
logS1
logS0
6
4
2
0
-2
-4
-4
-2
0
2
4
logS1 = logp1 - logp0 + logS0



Choose D=1
$\alpha_1$ = 0.3
Choose D=0
logS1
logS0
4
2
0
-2
-4
-2
0
2
4
logS1 = logp1 - logp0 + logS0

# Generalized Roy Model

- Generalized Roy model (Eisenhauer et al., 2015): Preference for alternative $d$ depends on $Y_{id}$ as well as a cost $C_{id}$:

$$D_i = 1\left\{Y_{i1} - C_{i1} > Y_{i0} - C_{i0}\right\}$$

- Allows us to ask richer questions about selection on both levels and gains
  - Is average $Y_{i1}$ higher or lower for individuals that choose $D_i = 1$?
  - Is average $Y_{i0}$ higher or lower for individuals that choose $D_i = 1$?
  - Are average *gains* $Y_{i1} - Y_{i0}$ larger or smaller for individuals that choose $D_i = 1$?
- Close link between generalized Roy model and econometric models of treatment effect heterogeneity

# Generalized Roy Model

- Write outcomes and costs as functions of characteristics $X_i$:

$$Y_{id} = \mu_d(X_i) + \epsilon_{id}, \;\; C_{id} = \varphi_d(X_i) + \eta_{id}, \;\; d \in \{0, 1\}$$

- $E[\epsilon_{id}|X_i] = E[\eta_{id}|X_i] = 0$ by definition of $\mu_d(\cdot)$ and $\varphi_d(\cdot)$
- Decision rule is

$$D_i = 1\left\{\mu_1(X_i) + \epsilon_{i1} - \varphi_1(X_i) - \eta_{i1} > \mu_0(X_i) + \epsilon_{i0} - \varphi_0(X_i) - \eta_{i0}\right\}$$

$$\implies D_i = 1\left\{\psi(X_i) > v_i\right\}$$

- Here $\psi(X_i) = [\mu_1(X_i) - \mu_0(X_i)] - [\varphi_1(X_i) - \varphi_0(X_i)]$, and $v_i = [\eta_{i1} - \eta_{i0}] - [\epsilon_{i1} - \epsilon_{i0}]$
- Dependence of decision on potential outcomes is embedded in $\psi(\cdot)$ and $v_i$

# Roy Model Identification

- Consider the regression of $Y_i$ on $X_i$ in the sample of individuals who chose $D_i = 1$
- The conditional expectation function is

$$E[Y_{i1}|X_i, D_i = 1] = \mu_1(X_i) + E[\epsilon_{i1}|X_i, D_i = 1]$$

$$= \mu_1(X_i) + E[\epsilon_{i1}|X_i, v_i < \psi(X_i)]$$

- Since $v_i$ includes $\epsilon_{i1}$, generally expect $E[\epsilon_{i1}|X_i, v_i < \psi(X_i)] \neq 0$, so observed mean outcome does not reflect population mean potential outcome $\mu_1(X_i)$
- Heckman (1974, 1976, 1979) idea: Work out the bias term to correct for self-selection

# Correcting Selection

- Suppose
$$(\epsilon_{id}, v_i)'|X_i \sim N\left(0, \left[\begin{smallmatrix} \sigma^2 & \rho_d\sigma_d \\ \rho_d\sigma_d & 1 \end{smallmatrix}\right]\right),\ d \in \{0,1\}$$
- This implies
$$\epsilon_{id} = \rho_d\sigma_d v_i + e_{id},\ E[e_{id}|v_i, X_i] = 0$$
- Conditional expectation function in selected sample is
$$E[Y_{i1}|X_i, D_i = 1] = \mu_1(X_i) + \rho_1\sigma_1 E[v_i|X_i, v_i < \psi(X_i)]$$
$$= \mu_1(X_i) + \rho_1\sigma_1\lambda_1(\psi(X_i))$$
- $\lambda_1(c) \equiv -\frac{\phi(c)}{\Phi(c)}$ is the inverse Mills ratio
- Likewise $E[Y_{i0}|X_i, D_i = 0] = \mu_0(X_i) + \rho_0\sigma_0\lambda_0(\psi(X_i))$, with $\lambda_0(c) = \phi(c)/[1 - \Phi(c)]$
- What restrictions on $(\rho_1\sigma_1, \rho_0\sigma_0)$ are implied by a pure Roy model?

# Generalized Roy Model Identification

- Two-step "Heckit" approach: Estimate $\psi(X_i)$ in a first-step probit, then include $\lambda_d(\hat{\psi}(X_i))$ in second-step OLS regression
- Mills ratio is a control function that adjusts for selection bias
- Problem: Dependent on functional form
    - If $X_i$ is a constant, cannot estimate
    - If $\mu(X_i)$ and $\psi(X_i)$ are unrestricted, cannot estimate
    - Estimable if we restrict $\mu_d(X_i)$ and $\psi(X_i)$ (e.g. make them linear) so that $\mu(X_i)$ is not colinear with $\lambda_d(\psi(X_i))$, but typically no good justification for this

# Generalized Roy Model Identification

- Solution: Suppose we have another variable $Z_i$ that affects costs but not potential outcomes:

$$E[Y_{id}|X_i, Z_i] = \mu_d(X_i), \ E[C_{id}|X_i, Z_i] = \varphi_d(X_i, Z_i)$$

- Participation rule becomes

$$D_i = 1\{\psi(X_i, Z_i) > v_i\}.$$

- Under normality, control function terms equal $\rho_d \sigma_d \lambda_d(\psi(X_i, Z_i))$, which vary conditional on $X_i$

- Sound familiar?

# Mulligan and Rubinstein (2008)

- Mulligan and Rubinstein (QJE 2008) apply the logic of self-selection models to study female wages and labor force participation
- Since the 1970s, female labor force participation has grown dramatically (from 46% in 1975 to 59% in 2007)
- The male/female wage gap has fallen, but within-gender inequality has increased for females
- MR hope to make sense of this by studying changes in the pattern of selection into the labor market
- Idea: Growing returns to human capital may produce more positive selection bias, closing the gender wage gap but increasing within-gender differences

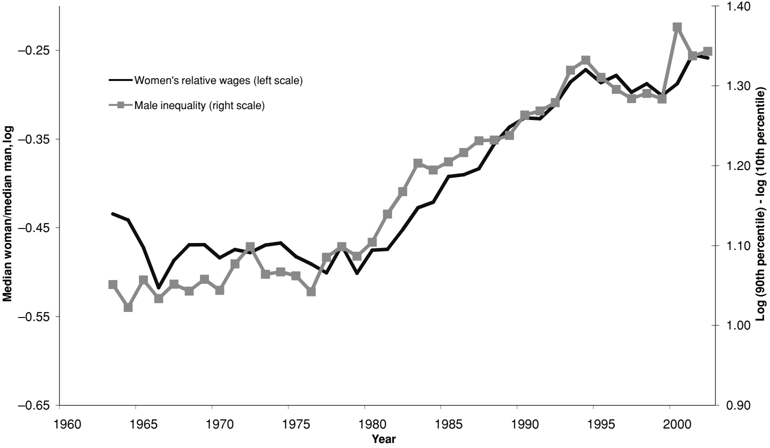



FIGURE I

Wage Inequality within and between Genders

The figure graphs time series of (a) the log of the ratio of the wage of the median working woman to that of the median working man (left scale, no markers), and (b) the log of the ratio of the wage of a man at the 90th percentile of the male wage distribution to that of a man at the 10th percentile (right scale, square markers). The calculations use our CPS wage sample of white persons aged 25–54, without trimming of outliers or adjusting topcodes.

# Mulligan and Rubinstein (2008)

- MR begin with the potential log wage equation

$$w_{it} = \mu_t^w + g_i \gamma_t + \sigma_t^w \epsilon_{it}^w$$

- $\mu_t^w$ is the overall mean potential log wage at time $t$
- $\sigma_t^w$ is the residual standard deviation of wages at time $t$
- $\gamma_t$ is the mean gender *potential* wage gap
- *Observed* wage gap $G_t$ (assume all men work):

$$\begin{aligned} G_t &= E[w_{it} | g_i = 1, L_{it} = 1] - E[w_{it} | g_i = 0] \\ &= \gamma_t + \sigma_t^w E[\epsilon_{it} | g_i = 1, L_{it} = 1] \\ &= \gamma_t + \sigma_t^w b_t. \end{aligned}$$

# Mulligan and Rubinstein (2008)

- Change in measured wage gaps:

$$\Delta G_t = \Delta\gamma_t + b_{t-1}\Delta\sigma_t^w + \sigma_t^w \Delta b_t$$

- Components due to change in potential wage gap, change in inequality for given selection bias, change in selection bias
- MR want to estimate the third term

# Mulligan and Rubinstein (2008)

- Participation determined by reservation wage $r_{it}$:

$$r_{it} = \mu_t^r + \sigma_t^r \epsilon_{it}^r$$

- Work decision is $L_{it} = 1\left\{r_{it} < w_{it}\right\}$, so

$$L_{it} = 1\{\epsilon_{it}^r - \tfrac{\sigma_t^w}{\sigma_t^r}\epsilon_{it}^w < \tfrac{\gamma_t + \mu_t^w - \mu_t^r}{\sigma_t^r}\}$$

- Behavioral change driven by factors common to all women (right-hand side), or factors that interact with women's characteristics (left-hand side)
- With small enough $\sigma_t^w$ and $\mathrm{Cov}(\epsilon_{it}^r, \epsilon_{it}^w) > 0$, working women will have lower potential wages (selection driven by low reservation wages)
- As $\sigma_t^w$ increases, participation increasingly driven by high wage offers $\Longrightarrow$ bias more positive

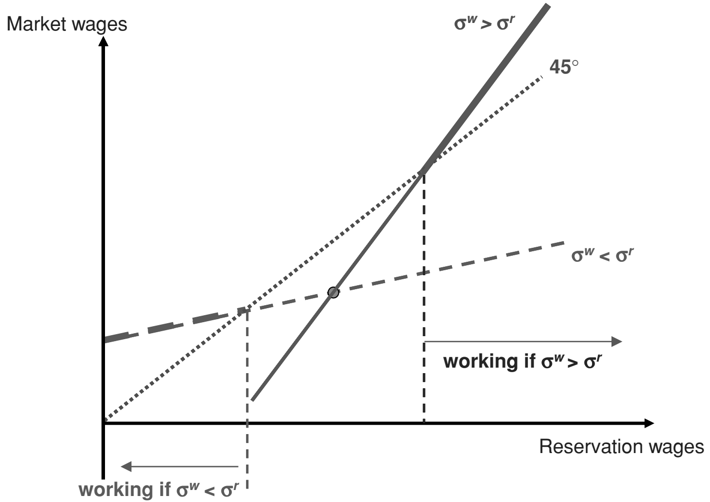



FIGURE II

GHR Model: Inequality Has Composition Effects on Measured Wages

The figure illustrates a comparative static for the Gronau-Heckman-Roy model with respect to $\sigma^w$; $\sigma^w$ ($\sigma^r$) denotes the standard deviation of market (reservation) wages. The 45-degree line partitions market workers from nonworkers. Two model parameterizations are shown: $\sigma^w$ greater (solid line) and $\sigma^r$ greater (dashed line). The thick portions of the line are above the 45-degree line and indicate workers.

# Mulligan and Rubinstein (2008)

- Under normality selection bias can be written

$$b_t = \theta \left( \sigma_t^w / \sigma_t^r \right) \lambda \left( P_t \right)$$

- $P_t$ is the fraction of women who work; $\sigma_t^w / \sigma_t^r$ is the relative importance of market wages in the participation decision
- Changes in bias can come from either source
- For a fixed selection rule, increasing $P_t$ reduces bias; but an increase in $\sigma_t^w$ changes the selection rule, possibly increasing bias

# Mulligan and Rubinstein (2008)

- MR assess changes in selection using the March CPS
- Two-step approach:

$$w_{it} = X_{it}\beta_t + g_i\gamma_t + g_i\theta_t\lambda(Z_{it}\delta_t) + u_{it}$$

- $X$: Education, marital status, potential experience, region
- $Z$: Same as $X$, plus number of young children...

TABLE I

CORRECTING THE GENDER WAGE GAP USING THE HECKMAN TWO-STEP ESTIMATOR

| Period | Method: OLS | Method: Two-Step | Bias |
|---|---|---|---|
| | Panel A: Variable Weights | | |
| 1975–1979 | −0.414 | −0.337 | −0.077 |
| | (0.003) | (0.014) | (0.015) |
| 1995–1999 | −0.254 | −0.339 | 0.085 |
| | (0.003) | (0.014) | (0.015) |
| Change | 0.160 | −0.002 | 0.162 |
| | (0.005) | (0.020) | (0.021) |
| | Panel B: Fixed Weights | | |
| 1975–1979 | −0.404 | −0.330 | −0.075 |
| | (0.003) | (0.014) | (0.014) |
| 1995–1999 | −0.264 | −0.353 | 0.089 |
| | (0.004) | (0.015) | (0.016) |
| Change | 0.140 | −0.024 | 0.164 |
| | (0.005) | (0.021) | (0.021) |

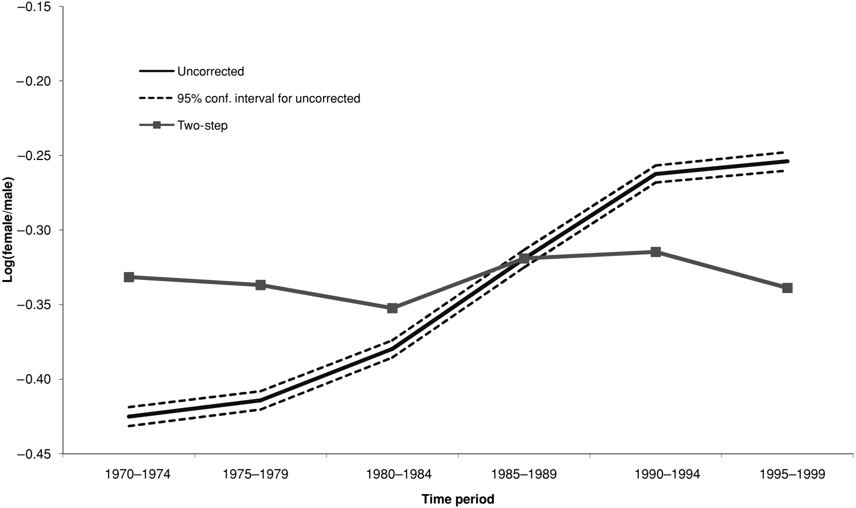



FIGURE III
Correcting the Gender Wage Gap: The Heckman Two-step Estimator

# Mulligan and Rubinstein (2008)

- The OLS gender wage gap fell by 16 log points from 1975-1995
- Controlling for the Mills ratio, the gap *widened*
- Two-step estimates suggest a reversal of bias, from negative in the 1970s to positive in the 1990s
- Taken at face value, the results imply that all of the decrease in the gender wage gap was due to changing selection

# Chandra and Staiger (2007)

- Chandra and Staiger (JPE 2007) consider self selection in a different context: treatment choice in health care
- Stylized fact: Large differences in medical treatments/spending across areas, without commensurate differences in outcomes (see Dartmouth map, or Atul Gawande)
- Flat of the curve explanation: High-use areas perform a lot of expensive treatments with low returns, so big gains could be had by cutting back
- CS ask whether this interpretation is compromised by self-selection

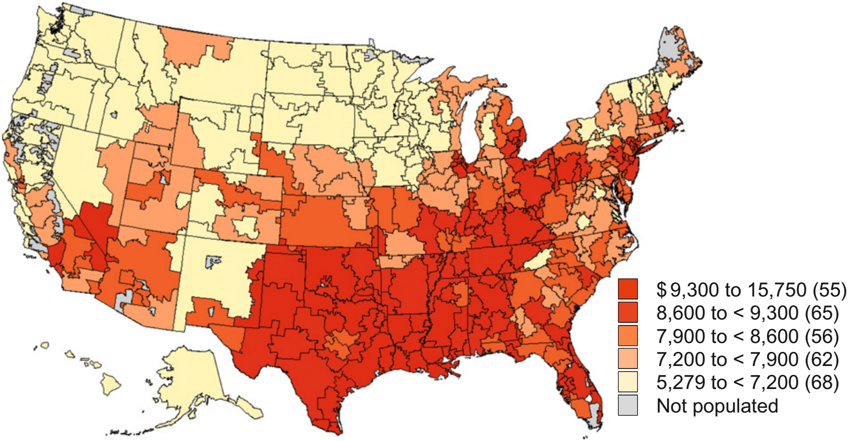



**Figure 2.4** Price-adjusted per capita medicare expenditures 2007. *Dartmouth atlas of healthcare.*

# Chandra and Staiger (2007)

- CH model: Treatment choice with productivity spillovers
- Consider the case of heart attacks (acute myocardial infarction, AMI)
- Two possible treatments:
  1. Non-intensive medical management
  2. Intensive intervention (surgery; bypass or angioplasty)
- Physician chooses treatment option $i \in \{1, 2\}$ that maximizes utility for each patient, which depends on survival probability $Survival_i$ and cost $Cost_i$
- Survival probability depends positively on population share receiving the same treatment, $P_i \implies$ positive productivity externality
- Justifications: Learning by doing ("see one, do one, teach one"), availability of support services for commonly used treatments, physician sorting

# Chandra and Staiger (2007)

- Formal Roy model: For $i \in \{1, 2\}$

$$Survival_i = \beta_i^s Z + \alpha_i^s P_i + \epsilon_i^s$$

$$Cost_i = \beta_i^c Z + \alpha_i^c P_i + \epsilon_i^c$$

$$U_i = Survival_i - \lambda Cost_i$$

$$= \beta_i Z + \alpha_i P_i + \epsilon_i$$

- Intensive treatment probability

$$Pr[i = 2] = Pr[U_2 > U_1]$$

$$= Pr[(\beta_2 - \beta_1)Z + \alpha_2 P_2 - \alpha_1(1 - P_2) + (\epsilon_2 - \epsilon_1) > 0]$$

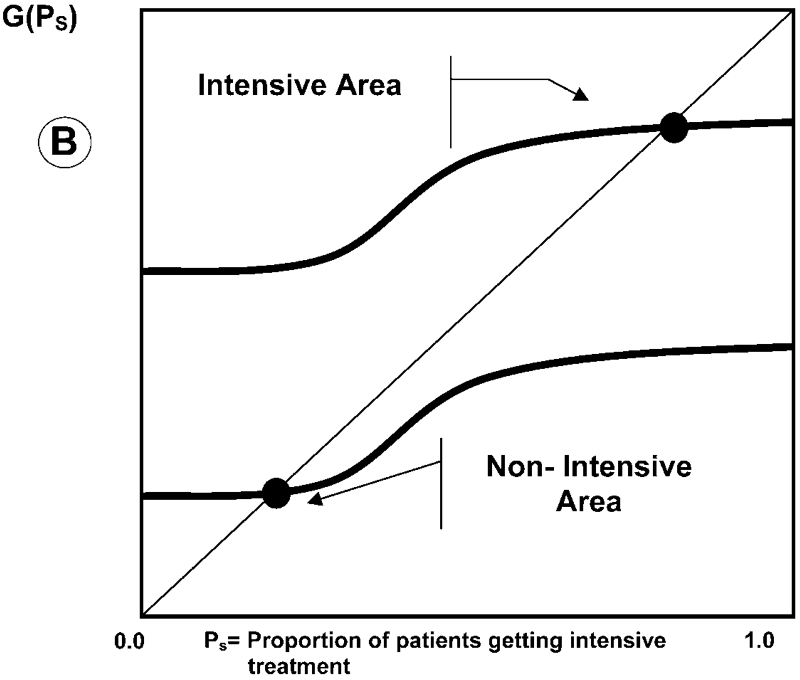
$$= Pr[\alpha P_2 - \alpha_1 + \beta Z > \epsilon]$$

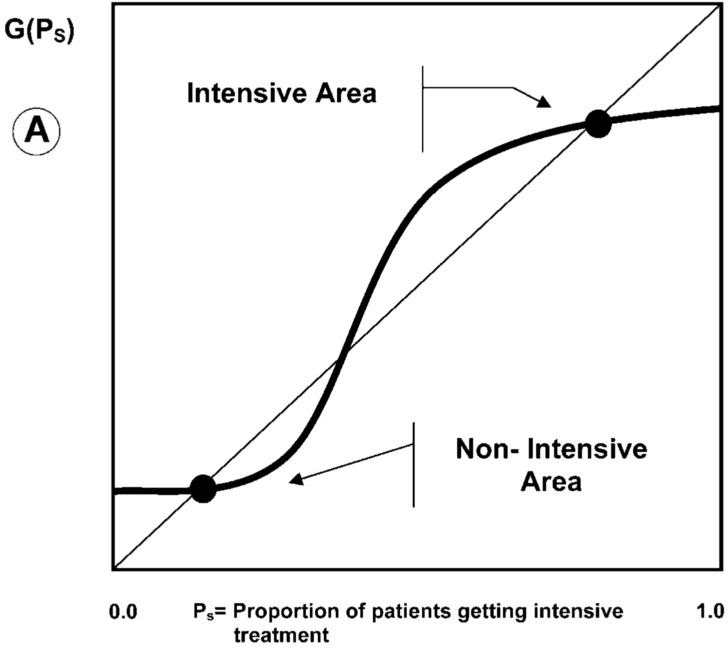
# Chandra and Staiger (2007)

- Equilibrium: $P_2$ must equal the fraction of individuals for whom treatment 2 is chosen (fixed point)

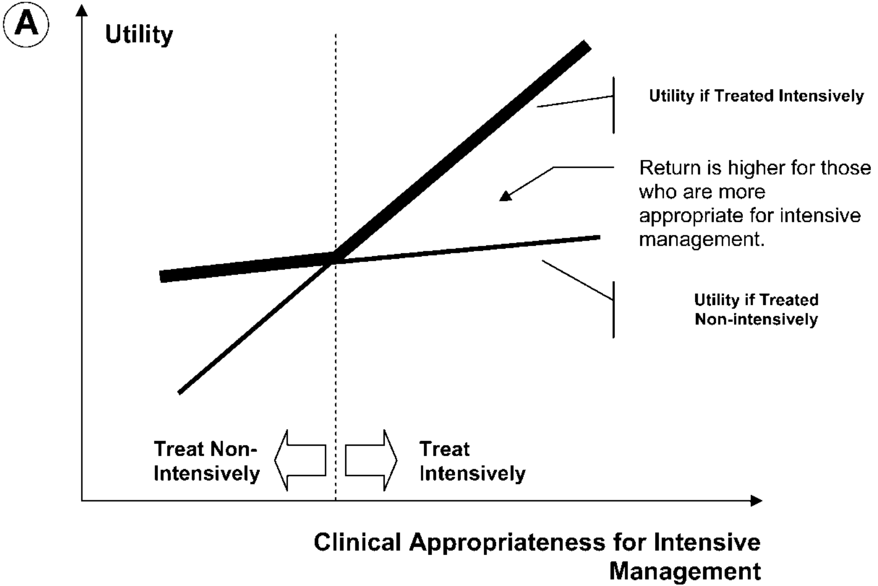
$$P_2 = \int Pr[\alpha P_2 - \alpha_1 + \beta Z > \epsilon] dF(Z)$$
$$= G(P_2)$$

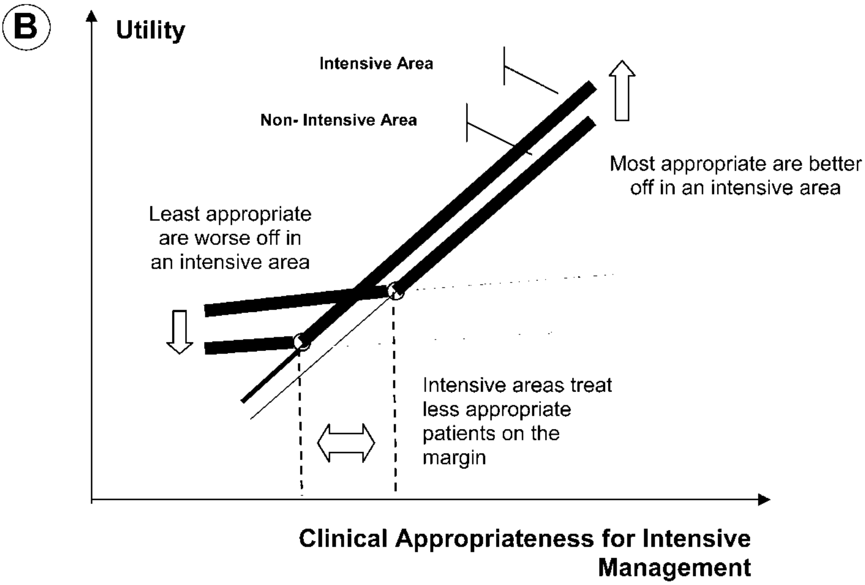
- Equilibrium proportion of intensive treatments must generate external benefits that justify treating this proportion intensively
- Produces the possibility of multiple equilibria


G(PS)
B
Intensive Area
Non- Intensive Area
0.0
1.0
Ps= Proportion of patients getting intensive treatment



$G(P_S)$
A
Intensive Area
Non- Intensive Area
0.0
$P_s$= Proportion of patients getting intensive treatment
1.0


A
Utility
Utility if Treated Intensively
Return is higher for those who are more appropriate for intensive management.
Utility if Treated Non-intensively
Treat Non-Intensively
Treat Intensively
Clinical Appropriateness for Intensive Management


B
Utility
Intensive Area
Non- Intensive Area
Most appropriate are better off in an intensive area
Least appropriate are worse off in an intensive area
Intensive areas treat less appropriate patients on the margin
Clinical Appropriateness for Intensive Management

# Chandra and Staiger (2007)

- Implications of the CS model:
  - Possibly multiple equilibria
  - Large differences in treatment rates can be generated by small differences in patient characteristics
  - Identical patients may be treated differently depending on population treatment rate
  - Gains to intensive treatment are larger for individuals who are most likely to be treated intensively ($TOT > ATE$)
  - Being in an intensive area is good for those most appropriate for the intensive treatment, and worse for those least appropriate
  - Marginal patient receiving the intensive treatment in an intensive area will be less appropriate for this treatment
  - For a given patient, treatment effect of intensive treatment is larger in an intensive area
- This model can explain why we might see large differences in treatment intensity without large differences in average outcomes

# Chandra and Staiger (2007)

- Welfare (expected utility): If differences in intensity are generated by multiple equilibria w/identical patients, can rank equilibria by computing costs and outcomes
- In this case, social planner will want to reduce variation in intensity (switch to less intense equilibrium) if intense areas have higher costs and no better outcomes
- If there is a single equilibrium with intensity that varies because of differences in patient characteristics, *not enough* variation in intensity – social planner would be willing to harm marginal patient to increase utility of inframarginal patients, and by assumption, decisionmaker (physician/patient) is not.

# Chandra and Staiger (2007)

- CS try to test the implications of their model
- Focus on heart attacks. Appropriate because:
  - A leading cause of death in the US
  - Geographically distinct markets
  - Two starkly different treatments: Intensive surgery (bypass), non-intensive management (thrombolytics)
  - Large differences across markets in intensity, but also variation in appropriateness across patients
- CS look at the effects of intensive treatment using data from the Cooperative Cardiovascular Project (CCP), a data set collected to try to analyze quality of heart attack treatment

# Chandra and Staiger (2007)

- Model for intensive treatment receipt:

$$Pr[\text{Cardiac Cath}_{ij} = 1 | X_i, j] = G(\theta_0 + \theta_j + X_i\Phi)$$

- Now $i$ is patient, $j$ is hospital referral region
- $X_i$ is detailed set of characteristics from patient chart
- Fitted value $\hat{G}(\theta_0 + X_i\Phi)$ captures appropriateness after taking out HRR effect
- $\theta_j$ captures differences in treatment rates for similar patients
- CS begin by testing for the presence of Roy selection: Are gains to intensive treatment increasing in the odds of getting this treatment?
- Model for effect of intensive treatment:

$$Y_{ij} = \beta_0 + \beta_1 \text{Cardiac Cath}_{ij} + X_i\Pi + u_{ijk}$$

- Instrument using differential distance to nearest cath hospital, splitting sample by $\hat{G}$

TABLE 2

Relationship between Differential Distance (DD) and Probability of Catheterization and Survival, and Differential Distance and Observable Characteristics (%)

| Sample | 30-Day Cath Rate: DD Below Median (1) | 30-Day Cath Rate: DD Above Median (2) | One-Year Survival: DD Below Median (3) | One-Year Survival: DD Above Median (4) | One-Year Predicted Survival: DD Below Median (5) | One-Year Predicted Survival: DD Above Median (6) | 30-Day Predicted Cath Rate for Patients Getting Cath: DD Below Median (7) | 30-Day Predicted Cath Rate for Patients Getting Cath: DD Above Median (8) |
|---|---|---|---|---|---|---|---|---|
| All patients ($N = 129{,}997$) | 48.9 | 42.8 | 67.6 | 66.7 | 67.5 | 67.2 | 63.3 | 63.2 |
| By cath propensity: | | | | | | | | |
| Above the median ($N = 64{,}733$) | 74.0 | 67.1 | 84.6 | 83.8 | 83.4 | 83.5 | 72.6 | 72.6 |
| Below the median ($N = 65{,}244$) | 22.9 | 19.5 | 50.1 | 50.4 | 51.1 | 51.6 | 32.3 | 32.5 |
| By age: | | | | | | | | |
| 65–80 ($N = 90{,}016$) | 61.1 | 54.9 | 74.3 | 73.5 | 73.9 | 73.9 | 67.4 | 67.3 |
| Over 80 ($N = 39{,}961$) | 20.3 | 16.5 | 52.1 | 52.1 | 52.6 | 52.7 | 34.6 | 34.1 |

Note.—Cath propensity is an empirical measure of patient appropriateness for intensive treatments. We define this measure by using fitted values from a logit model of the receipt of cardiac catheterization on all the CCP risk adjusters. Differential distance is measured as the distance between the patient's zip code of residence and the nearest catheterization hospital minus the distance to the nearest hospital.

TABLE 1

Instrumental Variable Estimates of Intensive Management and Spending on One-Year Survival by Clinical Appropriateness of Patient

| Sample | Instrumental Variable Estimates of: Impact of Cath On One-Year Survival (1) | Impact of Cath On One-Year Cost ($1,000s) (2) | Impact of $1,000 on One-Year Survival (3) |
|---|---|---|---|
| A. All patients ($N = 129{,}895$) | .142 | 9.086 | .016 |
| | (.036) | (1.810) | (.005) |
| B. By cath propensity: | | | |
| Above the median ($N =$ 64,799) | .184 | 4.793 | .038 |
| | (.034) | (1.997) | (.017) |
| Below the median ($N =$ 65,096) | .035 | 17.183 | .002 |
| | (.083) | (3.204) | (.005) |
| Difference | .149 | −12.39 | .036 |
| | (.090) | (3.775) | (.018) |
| C. By age: | | | |
| 65–80 ($N = 89{,}947$) | .171 | 6.993 | .024 |
| | (.037) | (1.993) | (.009) |
| Over 80 ($N = 39{,}948$) | .016 | 16.026 | .001 |
| | (.108) | (2.967) | (.007) |
| Difference | .155 | −9.033 | .023 |
| | (.114) | (3.574) | (.011) |

Note.—Cath propensity is an empirical measure of patient appropriateness for intensive treatments. We define this measure by using fitted values from a logit model of the receipt of cardiac catheterization on all the CCP risk adjusters. Differential distance (measured as the distance between the patient's zip code of residence and the nearest catheterization hospital minus the distance to the nearest hospital) is the instrument. Each model includes all the CCP risk adjusters, and the standard errors are clustered at the level of each HRR.

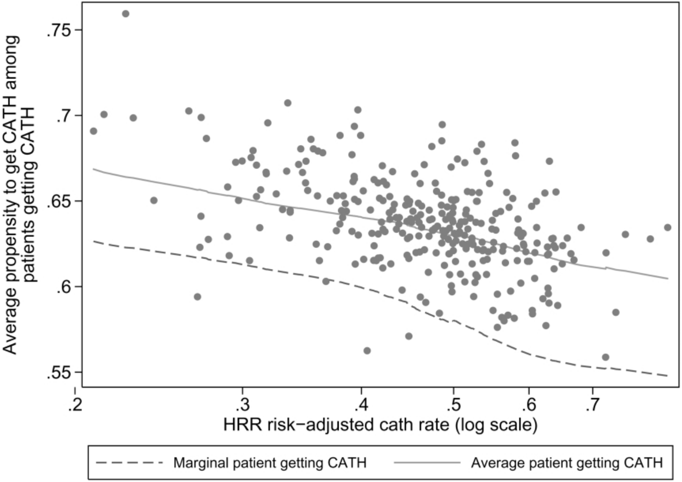



FIG. 3.—Relation between average patient and marginal patient receiving cardiac catheterization. For each of the 306 HRRs we graph the average propensity to receive cardiac catheterization (among patients who actually received it) against the log of the area risk-adjusted cath rate. Using local regression, we estimated the relationship between the average propensity and the risk-adjusted cath rate and the slope of this line at each point. These estimates were then used to plot the average (upper line) and marginal patient (lower line and estimated as the local difference in the average) receiving treatment.

# Chandra and Staiger (2007)

- Next, look for evidence of productivity spillovers
  - Is the quality of medical management worse in areas that do less of it?
  - Are identical individuals more likely to get intensive treatment in areas where other patients are more appropriate for it?
  - Is the return to intensive treatment higher in higher-intensity areas? Roy model with spillovers says yes – flat-of-the-curve explanation says no
  - Are the most-appropriate patients better off in high-intensity regions? Are the least-appropriate patients worse off?

TABLE 4

HRR-Level Measures of Intensive Treatment, Medical Management, Support of Medical Treatment, and Demographic Characteristics

| HRR Indicator | Mean | Standard Deviation | 10th Percentile | 90th Percentile | Correlation with HRR Cath Rate |
|---|---|---|---|---|---|
| Measures of intensive treatment: | | | | | |
| Risk-adjusted 30-day cath rate | 46.3% | 9.1% | 34.5% | 58.3% | 1.00 |
| Risk-adjusted 30-day PTCA rate | 17.7% | 5.1% | 11.3% | 23.6% | .81 |
| Risk-adjusted 30-day CABG rate | 13.4% | 2.9% | 10.2% | 16.9% | .51 |
| Risk-adjusted 12-hour PTCA rate | 2.7% | 2.6% | .6% | 5.8% | .52 |
| Measures of quality of medical management: | | | | | |
| Risk-adjusted beta-blocker rate | 45.6% | 9.5% | 34.2% | 58.3% | −.31 |
| Support for intensive treatment: | | | | | |
| Cardiovascular surgeons per 100,000 | 1.06 | .27 | .70 | 1.40 | .33 |
| Cath labs per 10,000 | 2.40 | .76 | 1.50 | 3.30 | .39 |
| Demographic characteristics: | | | | | |
| Log of resident population | 13.96 | .89 | 12.72 | 15.18 | −.05 |
| Log of per capita income | 9.55 | .20 | 9.31 | 9.85 | .02 |
| Percent college graduates | 19.3% | 5.5% | 13.1% | 26.6% | −.05 |

Note.—HRR surgical and medical intensity rates are computed as the risk-adjusted fixed effects from a patient-level regression of the receipt of cath or beta-blockers on HRR fixed effects and CCP risk adjusters.

TABLE 6
INSTRUMENTAL VARIABLE ESTIMATES OF INTENSIVE MANAGEMENT AND SPENDING ON SURVIVAL, BY SURGICAL INTENSITY OF HOSPITAL REFERRAL REGION

| Sample | Instrumental Variable Estimates of: Impact of Cath On One-Year Survival (1) | Impact of Cath On One-Year Cost ($1,000s) (2) | Impact of $1,000 on One-Year Survival (3) |
|---|---|---|---|
| A. All patients: | | | |
| HRR risk-adjusted cath rate: | | | |
| Above the median ($N$ = 63,771) | .256 | 6.691 | .038 |
| | (.061) | (3.510) | (.021) |
| Below the median ($N$ = 66,124) | .09 | 9.835 | .009 |
| | (.059) | (3.155) | (.007) |
| Difference | .166 | −3.144 | .029 |
| | (.085) | (4.720) | (.022) |
| B. Patients above the median cath propensity: | | | |
| HRR risk-adjusted cath rate: | | | |
| Above the median ($N$ = 32,388) | .271 | .347 | .78 |
| | (.064) | (4.370) | (9.820) |
| Below the median ($N$ = 32,411) | .168 | 4.962 | .034 |
| | (.046) | (2.890) | (.021) |
| C. Patients below the median cath propensity: | | | |
| HRR risk-adjusted cath rate: | | | |
| Above the median ($N$ = 31,383) | .206 | 16.21 | .013 |
| | (.129) | (5.130) | (.009) |
| Below the median ($N$ = 33,713) | −.139 | 22.064 | −.006 |
| | (.165) | (6.870) | (.007) |

NOTE.—HRR intensity rates are computed as the risk-adjusted fixed effects from a patient-level regression of the receipt of cath on HRR fixed effects and CCP risk adjusters. Differential distance (measured as the distance between the patient's zip code of residence and the nearest catheterization hospital minus the distance to the nearest hospital) is the instrument. Each model includes all the CCP risk adjusters, and the standard errors are clustered at the level of each HRR.

TABLE 7

Relationship between HRR Catheterization Rate, Survival, and Costs, by Clinical Appropriateness for Intensive Management

| | OLS Estimates of the Relationship between HRR Risk-Adjusted Cath Rate and | | | |
|---|---|---|---|---|
| Sample | One-Year Survival (1) | One-Year Cost ($1,000s) (2) | Beta-Blocker in Hospital (3) | Catheterization within 30 Days (4) |
| A. All patients ($N =$ 138,873) | .007 | 8.093 | −.28 | .702 |
| | (.019) | (1.410) | (.073) | (.004) |
| B. By cath propensity: | | | | |
| Top tercile ($N =$ 46,287) | .052 | 10.012 | −.366 | .802 |
| | (.019) | (1.439) | (.073) | (.032) |
| Middle tercile ($N =$ 46,295) | .03 | 11.154 | −.271 | .906 |
| | (.030) | (1.784) | (.082) | (.021) |
| Bottom tercile ($N =$ 46,291) | −.075 | 2.763 | −.209 | .369 |
| | (.028) | (1.612) | (.073) | (.021) |
| Difference (top − bottom) | .127 | 7.249 | −.157 | .433 |
| | (.034) | (2.161) | (.103) | (.038) |
| C. By age: | | | | |
| 65–80 ($N = 96{,}093$) | .023 | 9.616 | −.311 | .775 |
| | (.021) | (1.448) | (.072) | (.012) |
| Over 80 ($N = 42{,}780$) | −.031 | 4.738 | −.215 | .531 |
| | (.028) | (1.603) | (.080) | (.022) |
| Difference (top − bottom) | .054 | 4.878 | −.096 | .244 |
| | (.035) | (2.160) | (.108) | (.025) |
| D. By AHA/ACC criterion: | | | | |
| Ideal ($N = 89{,}569$) | .027 | 9.845 | −.302 | .769 |
| | (.023) | (1.599) | (.076) | (.010) |
| Appropriate ($N =$ 31,800) | −.002 | 6.174 | −.282 | .752 |
| | (.024) | (1.537) | (.080) | (.026) |
| Not appropriate ($N =$ 17,504) | −.08 | 2.958 | −.177 | .264 |
| | (.040) | (1.511) | (.065) | (.021) |
| Difference (top − bottom) | .107 | 6.887 | −.125 | .505 |
| | (.046) | (2.200) | (.100) | (.023) |

Note.—Cath propensity is an empirical measure of patient appropriateness for intensive treatments. We define this measure by using fitted values from a logit model of the receipt of cardiac catheterization on all the CCP risk adjusters. HRR surgical and medical intensity rates are computed as the risk-adjusted fixed effects from a patient-level regression of the receipt of cath or beta-blockers on HRR fixed effects and CCP risk adjusters.

# Chandra and Staiger (2007)

- Many predictions of the CS model with spillovers have support in the data
- Suggests that differences in treatment intensity across areas may be due to optimal decision-making in the presence of productivity spillovers, rather than inefficient utilization of care
- At the same time, welfare implications unclear
    - If patients are identical across areas, larger costs with the same outcomes implies that expected utility could be increased by reducing treatment intensity, regardless of whether variation is caused by productivity spillovers/multiple equilibria or flat of the curve
    - If source of variation in intensity is patient heterogeneity, we need *more* specialization
- Not obvious how to test whether variation is due to single vs. multiple equilibria

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